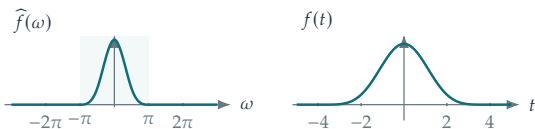


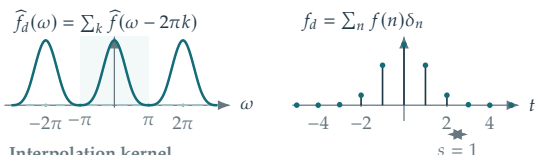
Shannon condition holds

$$f(t) = \text{sinc}^4(t/4)$$

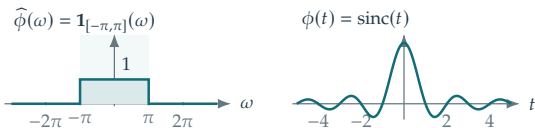
Original signal



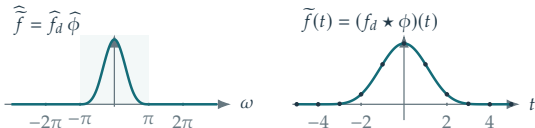
Sampling



Interpolation kernel



Reconstruction



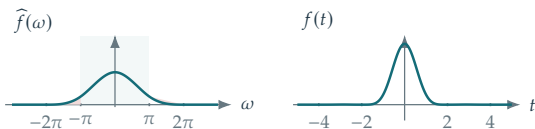
$\tilde{f} = f$: exact reconstruction

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}, \quad \tilde{f}(t) = \sum_{n \in \mathbb{Z}} f(n) \text{sinc}(t - n)$$

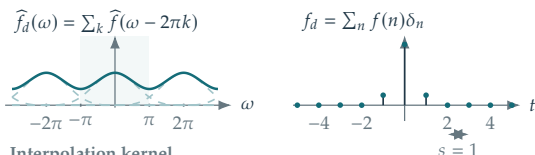
Shannon condition is violated

$$f(t) = \text{sinc}^4(t/2)$$

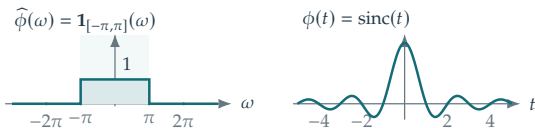
Original signal



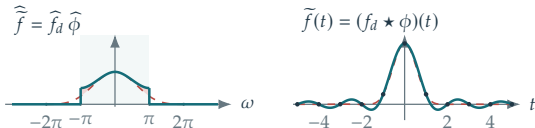
Sampling



Interpolation kernel



Reconstruction



$\tilde{f} \neq f$, although $\tilde{f}(n) = f(n)$

Pale band: $[-\pi, \pi]$. Dashed spectral curves: replicas. Red shading: aliased mass.
Dashed red in the reconstruction row: the original signal and spectrum. Dots: the shared samples.